saprophytes, cleistogamous flowers and extreme parasites endothecial differentiation is absent. The cells along the junction of the two sporangia of an anther lobe lack these thickenings. This region is called **stomium**. This region along with the hygroscopic nature of endothecium helps in the dehiscence of anther at maturity.

c. Middle layers: Two to three layers of cells next to endothecium constitute middle layers. They are generally ephemeral. They disintegrate or get crushed during maturity.

d. Tapetum: It is the innermost layer of anther wall and attains its maximum development at the tetrad stage of microsporogenesis. It is derived partly from the peripheral wall layer and partly from the connective tissue of the anther lining the anther locule. Thus, the tapetum is dual in origin. It nourishes the developing sporogenous tissue, microspore mother cells and microspores. The cells of the tapetum may remain uninucleate or may contain more than one nucleus or the nucleus may become polyploid. It also contributes to the wall materials, sporopollenin, pollenkitt, tryphine and number of proteins that control incompatibility reaction .Tapetum also controls the fertility or sterility of the microspores or pollen grains.

There are two types of tapetum based on its behaviour. They are:

Secretory tapetum (parietal/glandular/cellular): The tapetum retains the original position and cellular integrity and nourishes the developing microspores.

Invasive tapetum (periplasmodial): The cells loose their inner tangential and radial walls and the protoplast of all tapetal cells coalesces to form a periplasmodium.

Functions of Tapetum:

- It supplies nutrition to the developing microspores.
- It contributes sporopollenin through **ubisch bodies** thus plays an important role in pollen wall formation.
- The pollenkitt material is contributed by tapetal cells and is later transferred to the pollen surface.
- Exine proteins responsible for '**rejection reaction**' of the stigma are present in the cavities of the exine. These proteins are derived from tapetal cells.



Many botanists speak of a third type of tapetum called amoeboid, where the cell wall is not lost. The cells protrude into the anther cavity through an amoeboid movement. This type is often associated with male sterility and should not be confused with periplasmodial type.

2. Anther Cavity : The anther cavity is filled with microspores in young stages or with pollen grains at maturity. The meiotic division of microspore mother cells gives rise to microspores which are haploid in nature.

3. Connective: It is the column of sterile tissue surrounded by the anther lobe. It possesses vascular tissues. It also contributes to the inner tapetum.

Microspores and pollen grains

Microspores are the immediate product of meiosis of the microspore mother cell whereas the pollen grain is derived from the microspore. The microspores have protoplast surrounded by a wall which is yet to be fully developed. The pollen protoplast consists of dense cytoplasm with a centrally located nucleus. The wall is differentiated into two layers, namely, inner layer called **intine** and outer layer called **exine**. Intine is thin, uniform and is made up of pectin, hemicellulose, cellulose and callose together with proteins. Exine is thick and is made up of cellulose, sporopollenin and pollenkitt. The exine is not uniform and is thin at certain areas. When these thin areas are small and round it is called germ pores or when elongated it is called furrows. It is associated with germination of pollen grains. The sporopollenin is generally

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